

AI as the Phantom Limb: The Asymmetry of Attribution in Human vs. AI Delegation

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Abstract

AI is reshaping workplace dynamics as people increasingly delegate tasks to intelligent assistants. Yet how AI delegates are perceived compared to human delegates—and how their performance and their received feedback shape perceptions—remains unclear. We conducted a 2×2 between-subject experiment where participants delegated a scheduling task to either a human or an AI agent, varying their competence (high vs. low) and valence of received feedback (positive vs. negative) toward their performance. Participants generally had higher trust in human assistants; yet a striking asymmetry emerged: when an AI assistant received negative feedback, participants felt the criticism as more self-directed—an “AI Phantom Limb” effect—whereas positive feedback transferred less. This asymmetry did not appear with human delegates. These findings highlight broader design implications, suggesting that AI delegation might blur the boundary between self and other. We also discuss how these findings extend theories of delegation and responsibility attribution to AI.

CCS Concepts

• Human-centered computing → Empirical studies in collaborative and social computing.

Keywords

Delegation, AI Agent, Human-AI Collaboration, Extended Self

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1 Introduction

AI systems have rapidly shifted from simple tools of automation to active collaborators [32, 75, 103], raising a critical question: do people fundamentally perceive them differently from human collaborators? Delegation, a central form of human collaboration, enables individuals to offload tasks and coordinate efforts with others. As AI systems increasingly gain agency and increasingly serve as delegates on behalf of people—drafting proposals, scheduling meetings, or negotiating agreements—understanding how people perceive these AI delegates and how they shape interpersonal dynamics has become increasingly critical.

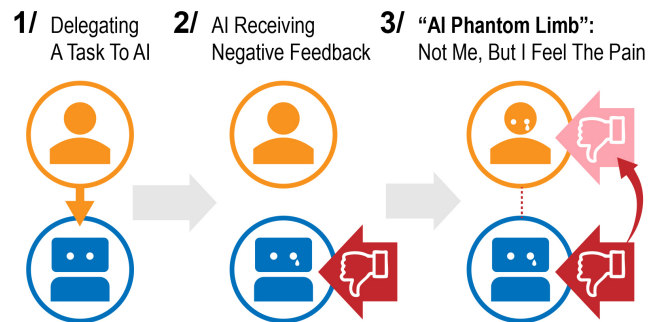


Figure 1: Delegation dynamics between a human delegator and an AI delegate. Delegators internalize feedback more strongly when negative feedback is directed at their AI delegates—an asymmetry not found with positive feedback or with human delegates.

Prior Human-Computer Interaction (HCI) research on delegation has primarily examined the competence and acceptance of AI systems and the conditions under which people decide to offload tasks, implicitly assuming that perceptions and underlying social mechanisms remain similar regardless of whether the delegate is human or AI. Although trust in AI has been studied and

compared against human collaborators [1, 3–5, 42, 70, 71, 102, 104], some important interpersonal dynamics taken for granted in Human–Human delegation—especially those related to the boundary between self and other, such as the attribution of responsibility, and the interpretation of feedback—have not been systematically investigated, even though they may be different in Human–AI interaction (HAI). Indeed, we have seen that the involvement of AI agents in collaboration has already produced tensions around self–other boundaries [22]. Research suggests that people often claim credit for positive outcomes produced by AI systems while shifting accountability for negative outcomes away from themselves, indicating a shift in responsibility attribution [12, 27, 30, 89]. Today, we are also seeing that many people are taking the credit for work done by AI and receiving compliments as though it were their own [26]. These patterns led us to question whether the clear-cut self–other boundaries found in Human–Human interaction may be fundamentally altered in HAI. Such dynamics are not entirely new in human society—for instance, supervisors often credit and receive compliments for subordinates’ good work while offloading responsibilities and blame to the subordinates when they do not do well. However, given how central these interpersonal dynamics are to collaboration and the fact that related concerns around the use of AI systems are growing, it is surprising that few systematic investigations have examined how these dynamics unfold differently in Human–AI versus traditional Human–Human delegation. Specifically, we would like to ask: Do people trust, perceive interpersonal distance, attribute responsibility, or internalize feedback differently in Human–AI versus traditional Human–Human delegation? And does the delegate’s competence or whether they receive good or bad feedback shape these interpersonal dynamics?

To address these questions, we conducted a $2 \times 2 \times 2$ between-participants experiment in which participants delegated a scheduling task to either a human or an AI assistant, systematically varying the type of the assistant (human vs. AI), competence (high vs. low), and the valence of feedback given to the assistant (positive vs. negative). We then measured trust, psychological distance, responsibility attribution, and the extent to which feedback directed toward the assistant was also perceived as directed toward the participant themselves.

This study makes two main contributions. First, we identify a striking asymmetry in delegation: participants internalized negative feedback directed at AI assistants as criticism of themselves. However, they did not internalize positive feedback to the same degree, and this asymmetry did not appear when the assistants were human—an effect we call “AI Phantom Limb” (see Fig. 1). Second, by comparing this finding with prior discussions on responsibility attribution in Human–AI interaction, we propose a more nuanced picture of how responsibility and accountability are perceived. When an AI performs poorly, participants may think the AI should bear responsibility; yet at the same time, they recognize that others may see the AI as ultimately unaccountable, causing criticism to be felt as directed toward themselves. This interweaving of responsibility and accountability reveals new complexities in how people experience collaboration with AI, and we conclude by discussing the theoretical, design, and social implications of these findings for AI collaborative systems in work settings.

2 Related Work

2.1 Delegation in Human–AI interaction

Delegation in Human–AI interaction refers to the process in which humans transfer task execution authority and, to some extent, outcome responsibility to AI agents [36, 66, 83]. This process enables humans to leverage AI’s computational efficiency while managing their own cognitive resources and decision-making load, particularly in complex or high-stakes tasks [19, 79, 91, 104]. For example, studies on Human–AI collaboration in decision-making contexts show that participants are more likely to delegate tasks involving large amounts of data processing or cognitively taxing choices, when they trust the AI’s competence [103].

Principal-agent theory provides a foundational framework to analyze delegation by highlighting the structural dynamics of authority, accountability, and information asymmetry between delegators and delegates [35]. Within this framework, delegation shifts authority and accountability, influencing human perceptions of control, risk, and task retention decisions [12, 29, 94]. Building on this, the task delegability framework further specifies that delegation decisions are contingent on both task- and agent-specific characteristics. Task attributes such as complexity, uncertainty, and potential consequences of errors consistently predict delegation behavior: humans tend to delegate tasks perceived as cognitively demanding or emotionally burdensome, while retaining control over tasks they perceive as critical or highly uncertain [14, 35, 67].

Moreover, considerations of task attributes can lead to more nuanced strategies. For instance, Lai et al. [54], in a case study on content moderation, found that humans adopt conditional delegation strategies when collaborating with AI, deciding whether to assign a task to the AI or retain control themselves based on specific task characteristics. Equally important in the delegation process is the type of agent involved. Empirical studies suggest that humans generally show higher willingness to delegate tasks to other humans than to AI agents [40]. This pattern appears to stem from social and cognitive mechanisms: people are more likely to attribute responsibility, predict behavior accurately, and perceive accountability in humans than in AI systems [100].

This reluctance arises not only from trust issues regarding AI competence but also from potential social consequences. For example, Sarkar [86] noted that “AI Could Have Written This” has become a classist slur in knowledge work, reflecting the social stigma associated with AI-assisted output. Consequently, Zhang et al. [105] observed that, to avoid such stigma, users sometimes engage in the secret use of large language models (LLMs) within their workflows.

The delegation process is not a one-time decision but a continuous cycle of trust, action, and evaluation [52]. After a task is delegated, the outcomes are rarely viewed in isolation; they are subject to assessment. Crucially, in organizational and collaborative settings, this assessment often comes from external sources. Research suggests that feedback serves as a primary mechanism for updating one’s mental model of a partner’s competence [21, 96, 105]. Especially in Human–AI collaboration, where trust is often fragile [24], understanding how these external evaluations influence the delegator’s perception is vital. This brings us to the specific

roles of trust, psychological distance, and feedback in shaping these dynamics, which we discuss in the following sections.

2.2 Trust and Psychological Distance in Human-AI Interaction

Trust is a cornerstone of effective HAI, defined as a user's willingness to be vulnerable to an agent's actions based on the belief that the agent is capable, reliable, and well-intentioned [55, 59]. Moreover, Liu [64] emphasized that an agent's agency locus and transparency are critical factors for reducing uncertainty and establishing trust. Recent HCI research has also highlighted the importance of trust calibration, as humans may overestimate or underestimate an AI's true capabilities, which in turn affects effective collaboration [17, 23, 53, 58, 68, 102].

Drawing on established theoretical frameworks in the Human-AI trust literature, this study treats trust as a multidimensional construct [71, 72, 82]. This framework can be traced back to the integrative organizational trust model proposed by Mayer, Davis, and Schoorman [70], which identifies three core foundations of trust: ability (competence), benevolence, and integrity. Subsequent research in HCI and automation trust has extended this framework, focusing on the dynamic and collaborative nature of the human-agent relationship. This contextualization led to the emphasis of four primary determinants: ability, benevolence, reciprocity, and perceived risk, which is a key determinant in trust calibration. [39, 42, 52, 87]

Synthesizing these contextualized models, this study adopts these four core elements commonly agreed upon in human-AI trust research: ability, benevolence, perceived risk, and reciprocity. Ability refers to users' assessment of the agent's technical performance and reliability [104]; benevolence reflects positive judgments of the agent's intentions, ensuring the AI operates in the user's best interest [57]; perceived risk represents the potential negative consequences users feel when relying on the AI, particularly under uncertainty [46] and reciprocity is defined as the user's belief that the agent will provide commensurate benefits or return the favor in an exchange relationship, a foundational social dynamic in HCI established by Fogg and Nass [34].

While trust addresses the evaluative aspect of the interaction, the perceived closeness between a human and an AI agent is equally crucial for successful collaboration. This perception is grounded in the concept of psychological distance from Construal Level Theory (CLT) [99]. CLT posits that psychological distance comprises four key dimensions: temporal, spatial, social, and hypothetical. In the context of HAI, this study examines psychological distance primarily through its social dimension—often operationalized as social distance—which reflects the extent to which humans perceive the AI agent as socially present or human-like [49]. Kim & Mutlu found that this social proximity is a key factor influencing Human-AI interaction.

This sense of closeness or remoteness is largely shaped by users' mind perception of the AI agent, with the ascription of mental states leading to reduced psychological distance [101]. Within this framework, psychological distance reflects the degree to which humans perceive AI agents as possessing social characteristics, such as intentionality, responsiveness, and human-like behaviors.

Reduced psychological distance (i.e., feeling closer via social cues) facilitates a more cooperative human-AI relationship and makes users more willing to rely on the AI for task execution [59, 81].

Moreover, the AI's behavioral style also affects user reliance. Xu, Lien, and Höllerer [103] reported that when the AI adopts a more zealous recommendation strategy, users are more likely to follow its suggestions, indicating that the interaction effect between competence and psychological distance may be more complex than previously assumed.

Building on these insights, psychological distance can be considered a relevant social-contextual factor that may influence delegation intention and behavioral reliance on AI agents. Prior research suggests that social coordination and interactional synchrony can enhance willingness to collaborate in human-automation or human-robot collaboration [77]. For instance, Bartkowski et al. [11] found that movement synchronization with a non-humanoid robot increased participants' trust in the robot, while Schneiders et al. [88] showed that temporal entrainment in collaborative tasks positively affected trust and willingness to collaborate in dyads. Collectively, these studies highlight the perception of distance as an important variable shaping the user's decision to delegate tasks. Accordingly, in this study, we treat psychological distance as a relevant social-contextual variable, examining its potential influence on responsibility attribution and delegation intention.

2.3 Feedback in Human-AI delegation

In delegation contexts, external feedback serves as a critical social factor. Drawing from foundational organizational research [7, 44], feedback is not merely a task metric but a social evaluation representing whether the agent has met external expectations and norms [6, 90]. This external perspective is crucial because it introduces the broader social standards that underlie accountability assessments [65].

Regarding the impact of feedback valence, literature in organizational psychology and Human-Human interaction suggests that the nature of evaluation significantly directs the delegator's trust and coping mechanisms. Positive feedback typically acts as a social validation, reinforcing confidence in the delegate's competence and fostering goal commitment [8, 33]. Conversely, in interpersonal contexts, negative feedback signals a violation of expectations, often inducing social stress and emotional regulation efforts, which prompts the delegator to reassess their reliance on the agent [13, 51, 69, 95].

Recent HCI studies have demonstrated that these psychological dynamics extend to Human-AI delegation. Spitzer et al. [93] showed that contextual information and third-party assessments influence how delegators evaluate AI performance. Extending this, Ahn et al. [2] found that outcome feedback and model interpretability jointly affect users' trust in AI, highlighting that clear feedback from external sources can improve confidence in AI decision-making. Similarly, Thuillard et al. [97] demonstrated that negative feedback, even when directed at AI systems, induces social stress and alters subjective perceptions of competence, suggesting that the valence of feedback plays a critical role in trust dynamics similar to Human-Human interactions [60]. Xu et al. [103] further

revealed that expectation disconfirmation through feedback impacts trusting intentions in Human–AI teams. Collectively, these studies underscore that feedback directed toward AI—as a proxy for social judgment—can shape how delegators perceive competence, assign responsibility, and calibrate trust. In this study, we build on this literature to examine how trust, psychological distance, responsibility attribution, and external feedback interact when tasks are delegated to an AI agent.

2.4 Responsibilities Attribution

Attribution studies provide a common method of measuring responsibility in psychology [48]. Traditional attribution theory suggests that when behavior is consistent with prior expectations, people attribute causes of action to internal characteristics; conversely, when behavior is inconsistent with expectations, causes are attributed to external situational factors [15, 47]. Drawing on classic attribution theory—particularly how humans act as “intuitive psychologists” when interpreting others’ intentions—traditional Human-to-Human cooperation benefits from shared mind perception. As a result, attribution boundaries are often clear: actors can readily assign blame (e.g., to a subordinate) or claim credit for success [76, 80, 84].

However, in HAI, there is a more delicate relationship when the difference between humans and AI is added to the dynamics. The incorporation of AI, which blurs the self-other boundary due to varying levels of mind perception, makes the situation subtly more complex. Consequently, prior findings on responsibility attribution are often inconsistent. For example, it was found that in AI-mediated communication, the AI absorbs part of the responsibility when outcomes go wrong [43]. Nevertheless, Shank et al. [90] pointed out that when erroneous actions occur, although people do assign moral blame [56] to the AI entity, the degree of blame is often lower than that assigned to human agents, and people are more likely to shift ultimate moral responsibility to the humans who design or program the AI (e.g., AI system developers). This aligns with the findings of Tolmeijer et al. [98], who noted that users perceive AI as capable but not necessarily responsible, while developers (designers) are seen as partially accountable for moral responsibility.

These divergent outcomes stem from the fact that users’ internal judgments often conflict with external requirements for accountability [92]. When viewed solely from a social perspective on accountability in Human–AI collaboration, blame is typically directed toward the human rather than the system [6]. For instance, Lima et al. [63] found that external observers tend to assign more responsibility to human developers or operators rather than to the algorithmic system itself. This tendency is echoed in the legal attribution of fault to the human safety driver in the fatal Uber self-driving vehicle accident [30], in which Elish et al. conceptualized this as the “moral crumple zone,” a metaphor likening humans in Human–AI systems to the crumple zones in cars: just as the vehicle structure absorbs the force of impact in a crash, human operators absorb the burden of blame in the aftermath of system failures.

In sum, this study treats responsibility as an internal judgment—the user’s subjective evaluation of who should bear the causal or moral

burden for an outcome—whereas accountability refers to the external requirements for answerability imposed by social, legal, or organizational norms. This distinction is further elaborated in the discussion section.

3 Current Study

3.1 Trust In Delegation: The Role of Agent Type (AI vs. human), Competence, And Feedback

The concept of trust in delegation has been central in principal-agent theory, where the relationship between the delegator (principal) and the delegate (agent) is critical for successful task completion [35]. Previous studies suggest that trust in the agent is influenced by several factors, such as the agent’s competence and the type of feedback received. Specifically, trust in human agents is generally higher than in AI agents [55], and when the delegator perceives the agent as more capable of managing complex tasks.

The competence of the agent also plays a pivotal role in trust. Competence, referring to the agent’s ability to perform tasks effectively [9], is a well-established determinant of trust in both human and AI delegates [23, 39]. Research shows that delegators are more likely to trust competent agents, as they are perceived as more reliable and capable of delivering results [59]. Consequently, we hypothesize that delegators will experience higher trust when delegating tasks to humans compared to AI, and when delegating tasks to more competent agents.

Moreover, the feedback directed at the delegate received by the delegators plays an important role in shaping trust. Positive feedback, such as compliments or recognition, is likely to increase trust in the agent [62], while negative feedback, such as criticism, may diminish trust. The valence of feedback, therefore, is hypothesized to significantly affect trust in the delegation process. Thus, we hypothesize the following:

- *H1-1. Delegators will experience higher trust when delegating tasks to humans compared to AI.*
- *H1-2. Delegators will experience higher trust when delegating tasks to more competent delegates.*
- *H1-3. Delegators will experience higher trust when the feedback given to the delegate is positive.*

3.2 Psychological Distance by the Agent Type (AI vs. human), Competence, And Feedback

Psychological distance refers to the perceived emotional closeness or detachment between the delegator and the delegates [99]. In the context of delegation, a closer relationship (or lower psychological distance) tends to lead to better communication, increased cooperation, and a stronger sense of shared responsibility [16]. People typically feel closer emotional bonds with human agents than with AI [40], reflecting a lower psychological distance when working with a human compared to an AI agent. This aligns with findings from social psychology, where people generally form stronger emotional connections with human beings than with AI systems [85], which are often perceived as more mechanical and less relatable. We anticipate that this will also hold true when humans are delegating tasks to human delegates vs. AI delegates. The competence of the agent also affects psychological distance. Delegators may

experience lower psychological distance toward competent agents, as they are perceived as more capable of understanding and executing tasks effectively [73]. Conversely, delegating tasks to less competent agents may increase psychological distance, as the delegator may feel the need for closer supervision or increased caution [99]. The feedback given to the delegates (received by the delegator) can influence psychological distance. Positive feedback fosters a sense of accomplishment and strengthens the relationship between the delegators and the delegates, thereby reducing psychological distance [75]. In contrast, negative feedback may increase psychological distance, particularly when it implies failure or criticism [74]. Based on these considerations, we hypothesize the following:

- *H2-1. Delegators will experience lower psychological distance toward the delegates when the delegate is a human compared to an AI.*
- *H2-2. Delegators will experience lower psychological distance toward the delegates perceived as more competent.*
- *H2-3. Delegators will experience lower psychological distance toward the delegates receiving positive feedback.*

3.3 Research Questions: Responsibility Attribution and Feedback Direction

Given the findings in previous literature, we are also interested in the interaction between the type of the delegate (humans vs. AI) and the delegates' competence (high vs. low) and their received feedback (positive vs. negative) on trust and psychological distance. We therefore raise several research questions. Besides, given the delegators and delegates are usually perceived as functioning together as a team, we are also interested in the attribution of the feedback and the responsibilities. Therefore, we also investigate to what degree the delegators also feel like the feedback given to the delegates is also toward them, and how they attribute the responsibilities between all stakeholders including the delegators and delegates. We therefore raised the following research questions:

- *RQ1: Given the different agent types (human vs. AI) and competence (high vs. low) of delegates and their varying received feedback (positive vs. negative), how will the delegator perceive their delegate differently?*
- *RQ1-1: How do agent type, competence, and feedback interact to predict trust?*
- *RQ1-2: How do agent type, competence, and feedback interact to predict psychological distance?*
- *RQ1-3: How does the extent to which delegator perceives feedback directed at the delegate as also directed at themselves vary according to agent type, competence, and feedback?*
- *RQ1-4: How does the responsibility attribution vary according to agent type, competence, and feedback?*

4 METHODOLOGY

4.1 Participants

Following approval by the National Chengchi University Institutional Review Board (Protocol ID: NCCU-REC-202507-I136), we recruited 355 participants on Prolific with a compensation of £2.5 for each participant (190 males, 161 females, 4 prefer not to say;

age = 18–87, $M = 42.77$, $SD = 13.37$). Eligibility criteria required participants to be over 18 years old and located in English-speaking countries. We also asked participants to self-report their level of AI familiarity ($M = 3.77$, $SD = 0.97$; in the Likert scale of 5) and the frequency of AI usage (Several days a week: 22.3%, More than one time a week: 21.1%, and less than once a week: 19.7%).

4.2 Design

Our study investigates how people perceive AI vs. human delegates in workplace settings. In all conditions, each participant (who is the delegator) was paired with an assistant (the delegate) and a client to simulate a scenario in which the participant asked their assistant to schedule a meeting time with the client. We employed a between-subject $2 \times 2 \times 2$ experimental design—the type of the assistants (human vs. AI), competence of the assistant (high vs. low) and feedback valence given to the assistant by the client (positive vs. negative), yielding eight experimental conditions in total. The main part of the study unfolded across five phases. In the first phase (providing information), participants indicated their names and their most ideal and least ideal time slots for an online meeting next week while being informed about the scenario: “Imagine you are working in a consultancy agency. You need to schedule a one-hour online meeting with a very important client. Please check your actual schedule and identify the ideal time slots for this meeting.” We asked them to check their real schedule for next week so that the whole experiment felt realistic. This provided a solid basis for the following phases.

During the second phase (matching), participants were first randomly assigned to either a human (shown as “Your Human Assistant Alex”) or an AI assistant (shown as “Your AI assistant”). On the next page, the participants were then made to believe that they were connected to their client (shown as “Your Client Andy”), who is another Prolific user online (while in fact it is simulated and not a real person). To avoid confounds, assistants and clients were presented with a gender-neutral graph throughout the study. Adequate delays were introduced in the matching process to make the participants believe that the system was really connecting them to the assistant and the client.

In the third phase (negotiation), the participants were first told by the assistant, “Hi [NAME]! Now I am going to negotiate on your behalf”. An animation of chat bubbles was then shown to indicate that the assistant was negotiating a meeting time with the client. In this phase, assistant competence was manipulated through response latency: while it was shown to the participant that the average time for negotiation is one minute, the assistants in the high-competence condition completed the negotiation in 37 seconds, while low-competence assistants required 137 seconds. A ticking timer was always presented so that the participants could tell the time clearly.

In the fourth phase (outcome), participants were shown the final scheduled meeting time, as well as the time consumed in the negotiation process. We clearly noted whether it is “much faster than the average” or “much slower than the average”. The negotiation outcomes were also decided by the assistant's competence and the participants' pre-stated availability: high-competence assistants secured preferred time slots, while low-competence assistants

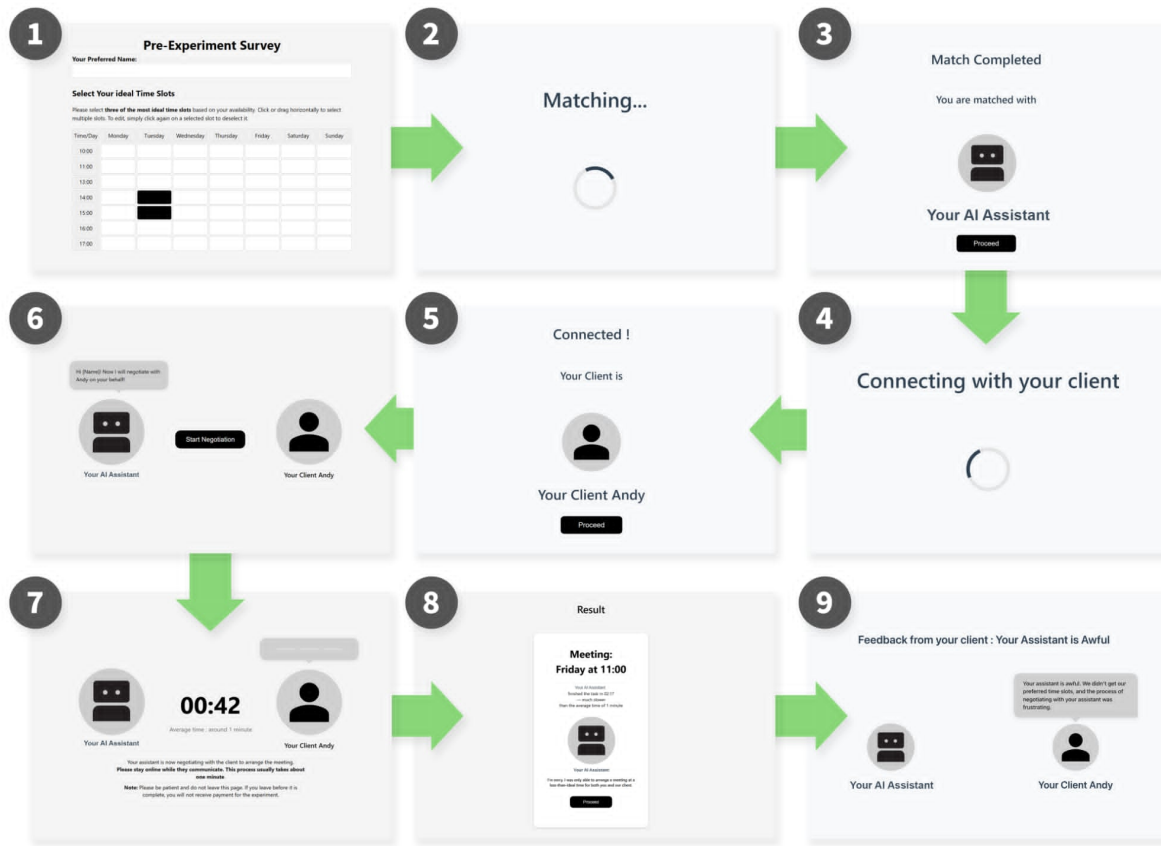


Figure 2: The main part of the experiment: (1) Participants gave consent and provide their preferred name and most/least ideal time slots. (2)(3) Participants were matched with an assistant (human or AI) and, (4)(5) a client, with delays to reinforce the realism of the matching process, while both are actually simulated by scripts (6) Participants were first greeted by their assigned assistant, who would do the negotiation with the client on the participant's behalf. (7) The assistant then engaged in a live negotiation while the participant observed the animation, during which competence was manipulated by varying the negotiation latency (high competence = faster, securing ideal slots; low competence = slower, securing least-ideal slots). (8) Participants received the outcome of the negotiation, including the agreed meeting time and the duration taken, explicitly compared against the average. (9) Finally, participants were presented with client feedback for the assistant that was either positive or negative. Across all stages, assistant and client behaviors were identical, with only the assistant type, competence, and feedback valence manipulated to create eight experimental conditions.

returned least time slots. To simplify the situation, the slots that high-competence assistants secured were shown as "preferred" for both the participant and the client, and the time slots that low-competence assistants secured were also "least preferred" for both the participant and the client.

Finally, in the fifth phase (feedback), participants received qualitative client feedback that was either positive (compliment: "Your assistant is amazing!") or negative (criticism: "Your assistant is awful!") depending on the condition. According to different combinations of competence and feedback valence, the description will be a bit different. For example, high-competence and positive-feedback is "Your assistant is amazing! We both got our preferred time slots, and the communication process with your assistant was smooth.", while high-competence and negative-feedback is "Your assistant

is awful. Even though we both got our preferred time slots, the process of negotiating with your assistant was frustrating.

Importantly, both the assistant and client behaviors were fully scripted and identical across conditions, with only the labels and timing manipulations varying. This setup ensured strict experimental control while preserving ecological validity.

4.3 Measure

We assessed four dependent variables in this study

- (1) Trust: A 12-item, five-point Likert-Scale based on the Human-Computer Trust Model (HCTM) [38], covering perceived risk, benevolence, competence, and reciprocity. Scores were averaged into an "Overall Trust" measure. In our scenario of delegating tasks, the sub-scales perceived risk ($M=8.09$,

$SD=3.38$) and benevolence ($M=9.45$, $SD=3.63$) are the key for decision making of delegation, and the subscales competence ($M=8.78$, $SD=4.21$) and reciprocity ($M=8.33$, $SD=3.73$) are the key to make people spend more time interacting with computing entities.

- (2) Psychological Distance: A three-item, five-point Likert scale adapted from prior work, capturing perceived relational distance from the assistant in the interaction. [61] ($M=6.78$, $SD=3.26$)
- (3) Responsibility Attribution: Participants were asked to allocate 100% of outcome responsibility across themselves ($M=22.58$, $SD=20.42$), the assistant ($M=50.92$, $SD=24.63$), and the client ($M=26.504$, $SD=17.144$) according to a previously used percentage-allocation method [43].
- (4) Self-Directed Feedback Perception: Besides responsibility attribution, we are also interested in how people feel about the feedback—whether it is toward themselves—which we believe to be highly relevant but not identical to responsibility attribution. We therefore created a single-item measure tailored for this study, a five-point Likert scale measure in which participants rated agreement with the statement: “I feel that the comment given by the client not only directs at my assistant but also at me” (1 = entirely toward the assistant, 5 = entirely toward myself). The item was developed based on the conceptualization of responsibility in delegation contexts, which views responsibility as the perceived locus of control following automated assistance. ($M=3.55$, $SD=1.17$)

4.4 Material and Procedure

The experiment was delivered through a custom web-based platform hosted on Vercel, and participants recruited through Prolific were redirected to this custom platform, where they first read through the experiment descriptions and provided informed consent and then completed the main part of this study as described in the Design section (providing information-matching-negotiation-outcome-feedback). This streamlined online procedure ensured standardized experimental conditions across all participants while maintaining ecological validity through a realistic task scenario that mirrors common workplace scheduling negotiations. After the main part, participants were directed to complete the post-task questionnaire hosted on google form about the dependent variables, attention check and manipulation check questions, and basic demographic information such as gender and age. Participants were debriefed about the deception at the end of the questionnaire and given a code which they must copy and paste into the Prolific system to show that they have completed the experiment.

5 Result

5.1 Preliminary

Out of the 435 participants that we recruited on Prolific (235 men, 197 women, 3 undisclosed; age = 18–87, $M=52.85$, $SD=13.253$), we retained data from 355 participants (190 men, 161 women, 4 undisclosed; age range = 18–87, $M=42.77$, $SD=13.37$). This is because we implemented two safeguards. First, participants were required to correctly identify their assigned assistant type and the timeslot

outcome (ideal vs. non-ideal) as the attention check. Those who got any of these two questions wrong will be disregarded. Second, we monitored session duration to confirm sustained engagement during the negotiation task. Most participants complete the task within 8–20 minutes ($M=11.9$, $SD=12.82$), and those who finished in 5 minutes were disregarded, resulting in the final 355 participants.

Manipulation Checks: Independent-samples t-tests confirmed the success of our manipulations of competence and feedback valence. We asked participants two manipulation check questions. For “How competent do you think your assistant was in helping you schedule your meeting?”, participants perceived assistants in the high-competence condition as significantly more competent than those in the low-competence condition, $t(29.7) = 5.43$, $p < .001$. And for “How satisfied was your client with your assistant?”, participant’s responses were also significantly different between conditions of positive vs. negative feedback, $t(30.7) = 12.17$, $p < .001$.

5.2 Trust

H1-1, H1-2, H1-3, and RQ1-1 are all about the effect of independent variables—including assistant type (AI vs. Human), competence (high vs. low), and feedback valence (positive vs. negative)—on trust, so we conducted a three-way ANOVA to examine these effects simultaneously... see the left of Table 1.

H1-1. Delegators (Participants) will experience higher trust when delegating tasks to humans compared to AI.

Based on the three-way ANOVA result, we found a significant main effect between the two assistant types, $F(1,347) = 21.472$, $p < .001$, $\eta_p^2 = .058$. The human assistants ($M=3.09$, $SD=1.21$) were more trusted than AI assistants ($M=2.69$, $SD=1.078$) in delegation. H1-1 was supported.

H1-2. Delegators (Participants) will experience higher trust when delegating tasks to more competent delegates.

Participants who delegated tasks to high-competent assistants ($M=3.57$, $SD=0.916$) demonstrated significantly higher trust compared to those who delegated to low-competent assistants ($M=2.22$, $SD=0.97$), $F(1,347) = 234.930$, $p < .001$, $\eta_p^2 = .403$. H1-2 was supported.

H1-3. Delegators (Participants) will experience higher trust when the feedback given to the delegate is positive.

The results revealed that participants whose assistant received positive feedback from the condition gave significantly higher trust score ($M=3.28$, $SD=1.10$) than negative feedback conditions ($M=2.50$, $SD=1.08$), $F(1,347) = 78.240$, $p < .001$, $\eta_p^2 = .184$. H1-3 was supported.

RQ1-1: How do assistant type, competence, and feedback interact to predict trust?

For RQ1-1, we found no statistically significant two-way interaction effect, nor was the three-way interaction significant, see Table 1.

Our trust measurement included four subscales, and the analyses of these subscales are attached in Appendices A.1 and A.2.

5.3 Psychological Distance

Similar to trust, H2-1, H2-2, H2-3, and RQ1-2 are all about the effect of independent variables on psychological distance, so we also analyze them altogether here by running a three-way ANOVA, see

Table 1: Three-way ANOVA Analysis of Trust (Left) and Psychological Distance (Right)

	df	F	Trust			df	F	Psychological Distance		
			p-value	η_p^2				p-value	η_p^2	
Assistant Type	1	21.472	<.001***	0.058	1	6.582	0.010**	0.014		
Competence level	1	234.930	<.001***	0.403	1	53.105	<.001***	0.117		
Feedback valence	1	78.240	<.001***	0.184	1	39.993	<.001***	0.088		
Assistant type X Competence	1	2.388	0.123	0.003	1	0.492	0.483	0.001		
Assistant type X Feedback valence	1	1.271	0.260	0.001	1	0.117	0.771	0.000		
Competence X Feedback valence	1	0.939	0.333	0.001	1	0.084	0.732	0.000		
Assistant Type X Competence X Feedback valence	1	1.036	0.309	0.003	1	3.111	0.078	0.006		

the right of Table 1. For clarity, we have reverse-coded the statistics so that the higher scores represent greater psychological distance.

H2-1. Delegators (Participants) will experience lower psychological distance toward the delegate when the delegate is a human compared to an AI.

Participants who delegated to different assistant type (AI vs. Human) was statistically significantly different in terms of the psychological distance, $F(1,347) = 6.582, p = .010, \eta_p^2 = .001$. People perceived a higher psychological distance to the AI assistants ($M = 3.87, SD = 1.09$) than to the human assistants ($M = 3.62, SD = 1.06$). H2-1 was supported.

H2-2. Delegators (Participants) will experience lower psychological distance toward the delegate perceived as more competent.

Participants perceived a lower psychological distance to high-competent assistants ($M = 3.37, SD = 1.09$) than low-competent assistants ($M = 4.11, SD = 0.94$), $F(1,347) = 53.105, p < .001, \eta_p^2 = .117$. H2-2 was supported.

H2-3. Delegators (Participants) will experience lower psychological distance toward the delegate receiving positive feedback.

Participants perceive lower psychological distance to delegates receiving positive feedback ($M = 3.42, SD = 1.08$) than those receiving negative feedback ($M = 4.07, SD = 0.99$), $F(1,347) = 39.993, p < .001, \eta_p^2 = .088$. H2-3 was supported as well.

RQ1-2: How do assistant type, competence, and feedback interact to predict psychological distance?

The three-way ANOVA showed that there was no statistically significant two-way or three-way interaction effects, see Table 1.

5.4 Attribution of Feedback

RQ1-3: How does the extent to which delegators (Participants) perceive feedback directed at the assistant as also directed at themselves vary according to assistant type, competence, and feedback?

A three-way ANOVA examined the effects of Assistant Type (AI vs. Human), Competence (high vs. low), and Feedback Valence (positive vs. negative) on Self-Directed Feedback Perception (see Table 2). Two significant two-way interactions emerged.

First, the interaction between assistant type and feedback valence was significant, $F(1, 347) = 4.79, p = .029, \eta_p^2 = .013$, see Fig.3 (Left). Simple-effects tests revealed that within the AI condition, participants perceived client feedback as more self-directed when it was negative ($M = 3.81, SD = 1.21$) compared to positive ($M = 3.36, SD = 1.14$), $F(1, 177) = 6.61, p = .011, \eta_p^2 = .036$. However, within the Human condition, no difference emerged between negative ($M = 3.47, SD = 1.13$) and positive feedback ($M = 3.55, SD = 1.15$),

$F(1, 174) = 0.21, p = .645, \eta_p^2 = .001$. These results indicate that participants uniquely internalized negative feedback toward AI assistants as directed at themselves, whereas this asymmetry was not observed with human assistants. We call this phenomenon “AI Phantom Limb” and will discuss in detail in the discussion section. Second, the interaction between competence and feedback valence was also significant, $F(1, 347) = 6.82, p = .009, \eta_p^2 = .019$, see Fig.3 (Right). Simple-effects tests showed that under positive feedback, participants reported higher self-directed feedback perception in the high-competence condition ($M = 3.70, SD = 1.10$) than in the low-competence condition ($M = 3.21, SD = 1.15$), $F(1, 175) = 8.40, p = .004, \eta_p^2 = .046$. Under negative feedback, however, no difference emerged between high competence ($M = 3.57, SD = 1.18$) and low competence assistants ($M = 3.71, SD = 1.18$), $F(1, 176) = 0.65, p = .421$. Looking at the reverse contrast, within the low-competence condition, participants reported higher self-directed feedback perception following negative feedback ($M = 3.71, SD = 1.18$) than positive feedback ($M = 3.21, SD = 1.15$), $F(1, 178) = 8.29, p = .004, \eta_p^2 = .045$. By contrast, within the high-competence condition, feedback valence did not significantly affect perception ($M_{negative} = 3.57, SD = 1.18$; $M_{positive} = 3.70, SD = 1.10$), $F(1, 173) = 0.59, p = .442$. Together, these findings indicate that competence moderated how feedback was internalized: participants were more sensitive to negative feedback when the assistant was low-competence, and they attributed more self-directed perception under positive feedback when the assistant was high-competence.

5.5 Comparison Between Responsibility Attribution to the Participant vs. the Assistant

RQ1-4: How does the responsibility attribution vary according to assistant type, competence, and feedback?

Responsibility attribution in our study originally involved three possible targets—the participant themselves, the assistant, and the client—with allocations required to sum to 100%. For cases where responses did not sum exactly to 100, values were normalized proportionally (e.g., a response of 3–1–1 was rescaled to 60%–20%–20%). To simplify the analysis and focus on the central research question, we report here the results for responsibility attributed to the participant and to the assistant, omitting responsibility assigned to the client. Please see Appendix A.3 for the analysis of the responsibility attribution to the client.

Table 2: Three-way ANOVA Analysis of Self-Directed Feedback Perception

	df	F	Trust	
			p-value	η_p^2
Assistant Type	1	0.426	0.514	0.001
Competence level	1	1.962	0.162	0.005
Feedback valence	1	2.355	0.125	0.006
Assistant type X Competence	1	0.318	0.573	0.000
Assistant type X Feedback valence	1	4.794	0.029*	0.013
Competence X Feedback valence	1	6.819	0.009**	0.019
Assistant Type X Competence X Feedback valence	1	0.01	0.920	0.000

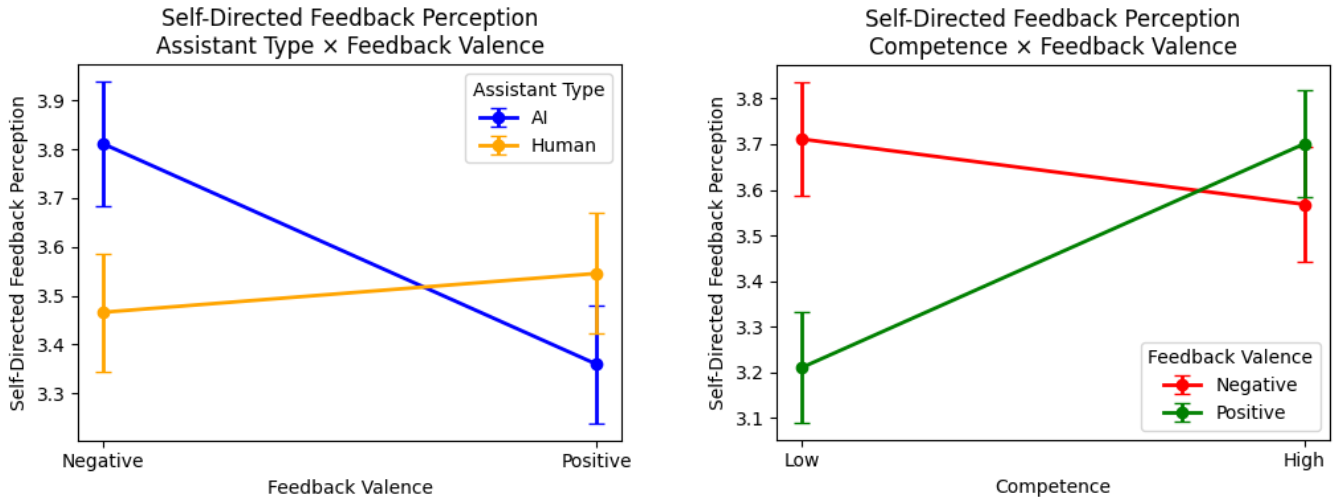


Figure 3: (Left) Interaction between assistant type and feedback valence on Self-Directed Feedback Perception: This plot illustrates the “AI Phantom Limb” effect: participants more likely to feel the feedback given to the assistant was also toward themselves when the feedback was negative and when the assistant was AI. **(Right) Interaction between competence and feedback valence on Self-Directed Feedback Perception.**

Table 3: Comparison Between Responsibility Attribution to the Participant vs. the Assistant

Assistant Type -Competence -Feedback Valence	N	Participant Mean (SD)	Assistant Mean (SD)	t	p-value
AI-Low-Negative	45	0.204 (0.227)	0.638 (0.283)	-5.99	<.001***
AI-Low-Positive	45	0.216 (0.213)	0.518 (0.242)	-4.71	<.001***
AI-High-Negative	45	0.249 (0.218)	0.384 (0.197)	-2.49	0.016*
AI-High-Positive	44	0.302 (0.158)	0.387 (0.206)	-1.69	0.098
Human-Low-Negative	45	0.170 (0.192)	0.638 (0.281)	-6.61	<.001***
Human-Low-Positive	45	0.168 (0.192)	0.528 (0.234)	-6.39	<.001***
Human-High- Negative	43	0.267 (0.197)	0.469 (0.196)	-3.54	0.001**
Human-High-Positive	43	0.234 (0.183)	0.507 (0.183)	-5.22	<.001***

We first conducted paired-samples t-tests within each condition (Assistant Type $\times$ Competence $\times$ Feedback), as reported in Table 3 including the means, standard deviations, and test results (The denotation “AI-High-Positive” means that it is an AI delegate which has high competence and it receives positive feedback). Across nearly all conditions, participants assigned significantly more responsibility to their assistants than to themselves, with particularly large

differences observed when the assistant was of low competence (e.g., AI-Low-Negative: $M_{\text{participant}} = 0.20$, $SD = 0.23$; $M_{\text{assistant}} = 0.64$, $SD = 0.28$; $t(44) = -5.99$, $p < .001$). The only exception was the AI-High-Positive condition, where the difference did not reach significance, $t(43) = -1.69$, $p = .098$. Overall, these results reveal a robust asymmetry: delegators consistently shifted responsibility to their assistants, distancing themselves particularly when assistants

Table 4: Three-way ANOVA Analysis of Responsibility Attribution to Assistant vs. Participant

	Attribution to Assistant			Attribution to Participant		
	F	p	η_p^2	F	p	η_p^2
Assistant Type	4.578	0.033**	0.013	2.403	0.121	0.006
Competence level	34.560	<.001***	0.090	11.756	<.001***	0.032
Feedback Valence	3.861	0.050	0.011	0.137	0.710	0.000
Assistant type X Competence	3.988	0.046**	0.011	0.141	0.707	0.000
Assistant type X Feedback valence	0.191	0.662	0.000	1.313	0.252	0.003
Competence X Feedback valence	7.584	0.006**	0.021	0.014	0.905	0.000
Assistant Type X Competence X Feedback Valence	0.062	.809	0.000	0.675	0.411	0.001

underperformed, regardless of whether the assistant was human or AI.

Then, two separate three-way ANOVAs were conducted with Assistant Type (AI vs. Human), Competence (High vs. Low), and Feedback Valence (positive vs. negative) as independent variables, and Responsibility Attribution to the Participant Themselves and Responsibility Attribution to The Assistant as dependent variables, respectively. Always use midrule to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily.

5.5.1 Responsibility Attribution to the Participant Themselves. A three-way ANOVA (see the right of Table 4) on responsibility attribution to the participant themselves revealed a significant main effect of competence, $F(1, 347) = 11.76, p < .001, \eta_p^2 = .033$. Participants attributed more responsibility to themselves when paired with high-competence assistants ($M = 0.26, SD = 0.19$) than with low-competence assistants ($M = 0.19, SD = 0.21$). No other main effects or interactions were significant.

5.5.2 Responsibility Attribution to The Assistant. A three-way ANOVA (See the left of Table 4) on responsibility attribution to the assistant revealed significant main effects of assistant type, $F(1, 347) = 4.58, p = .033, \eta_p^2 = .013$, competence, $F(1, 347) = 34.56, p < .001, \eta_p^2 = .091$, and feedback valence, $F(1, 347) = 3.86, p = .050, \eta_p^2 = .011$. Participants attributed more responsibility to human assistants ($M = 0.54, SD = 0.23$) than AI assistants ($M = 0.48, SD = 0.25$), more to low-competence assistants ($M = 0.58, SD = 0.26$) than high-competence assistants ($M = 0.44, SD = 0.20$), and marginally more to assistants receiving negative feedback ($M = 0.53, SD = 0.27$) than positive feedback ($M = 0.49, SD = 0.22$).

Two significant interactions further qualified these main effects. First, the Assistant Type $\times$ Competence interaction was significant, $F(1, 347) = 3.99, p = .047, \eta_p^2 = .011$, see Fig.4 (Left). Simple-effects analyses revealed that under high competence, participants attributed more responsibility to human assistants ($M = 0.49, SD = 0.19$) than AI assistants ($M = 0.39, SD = 0.20$), $F(1, 173) = 12.01, p < .001$. In contrast, no difference emerged under low competence ($M_{Human} = 0.58, SD = 0.26; M_{AI} = 0.58, SD = 0.27$), $F(1, 178) = 0.01, p = .915$. Conversely, competence effects held for both assistant types: participants attributed more responsibility to high-competence than low-competence assistants, both for AI assistants, $F(1, 177) = 29.54, p < .001$, and human assistants, $F(1, 174) = 7.44, p = .007$. Second,

the Competence $\times$ Feedback interaction was significant, $F(1, 347) = 7.58, p = .007, \eta_p^2 = .021$, see Fig.4 (Right). Simple-effects tests showed that under negative feedback, participants attributed more responsibility to low-competence assistants ($M = 0.64, SD = 0.28$) than to high-competence assistants ($M = 0.43, SD = 0.20$), $F(1, 176) = 33.66, p < .001$. Under positive feedback, the same pattern appeared though weaker: low-competence assistants ($M = 0.52, SD = 0.24$) received more responsibility than high-competence assistants ($M = 0.45, SD = 0.20$), $F(1, 175) = 5.34, p = .022$. Looking from the opposite perspective, feedback valence mattered only for low-competence assistants, who were attributed more responsibility under negative feedback ($M = 0.64, SD = 0.28$) than positive feedback ($M = 0.52, SD = 0.24$), $F(1, 178) = 8.81, p = .003$. No feedback effect emerged for high-competence assistants ($M_{negative} = 0.43, SD = 0.20; M_{positive} = 0.45, SD = 0.20$), $F(1, 173) = 0.47, p = .496$. In sum, participants consistently placed more responsibility on assistants than on themselves, which echoes previous research and will be discussed in depth in the next section. The only exception was when AI assistants were both competent and received positive feedback. We also found that participants tended to take more responsibility when paired with high-competence assistants, showing greater ownership when the delegate performed well. By contrast, assistants generally attracted more responsibility when they were humans, underperformed or received negative feedback. Together, these findings reveal the intricate dynamics of responsibility attribution in delegation.

6 Discussion

6.1 Main Findings

Our analyses revealed a consistent pattern across trust, psychological distance, feedback perception, and responsibility attribution when participants (delegators) delegate tasks to the assistants (delegates) while receiving feedback from third-party clients. For trust (H1-1 to H1-3), all hypotheses were supported. Participants reported higher trust when delegating to humans compared to AI (H1-1), when the delegate was high in competence (H1-2), and when the delegate received positive feedback (H1-3). However, for RQ1-1 we found no two-way or three-way interaction, suggesting that main effects dominated participants' judgments. Similarly, for psychological distance (H2-1 to H2-3), all hypotheses were supported. Participants experienced lower psychological distance toward human assistants (H2-1), toward high-competence assistants



Figure 4: (Left) The interaction between Assistant Type and Competence on Responsibility Attribution to Assistants: Participants attributed greater responsibility to assistants when their competence was low, with a steeper decline observed for AI assistants than for human assistants as competence increased. (Right) Interaction between Competence and Feedback Valence on Responsibility Attribution to Assistants: Negative feedback amplified responsibility attribution to low-competence assistants, whereas positive feedback led to a smaller difference across competence levels.

(H2-2), and in positive-feedback conditions (H2-3). Yet, for RQ1-2, no significant interactions were observed, again showing the dominance of main effects. By contrast, RQ1-3 uncovered critical interaction effect. Participants uniquely internalized negative feedback toward AI assistants as self-directed, while no such differences were found with positive feedback and with human assistant—a striking asymmetry we term the “AI Phantom Limb” effect. Moreover, competence interacted with feedback: under positive feedback, high-competence assistants elicited stronger self-directed perception than low-competence assistants, whereas under negative feedback, low-competence assistants triggered greater self-blame than high-competence ones. Finally, RQ1-4 examined responsibility attribution. Participants consistently assigned more responsibility to assistants than to themselves, particularly when assistants were human, underperforming, or negatively evaluated. On the other hand, the participants also took more responsibility when paired with high-competence assistants, suggesting that they were more willing to claim credit for successful outcomes achieved by capable delegates. In summary, all six hypotheses were supported, echoing previous research that people perceive other humans as more trustworthy and mentally closer than AI agents, with preference over individuals who are more competent and receive more positive feedback from others—we demonstrated that this also hold true in delegator-delegate relationship. In contrast, RQ1-3 and RQ1-4 revealed meaningful asymmetries, with negative feedback toward AI producing stronger self-directed costs and low-competence assistants disproportionately absorbing responsibility. Theoretical implications of these findings, as well as their design and broader social implications, are discussed in the following sections.

6.2 Theoretical Implications

6.2.1 AI Phantom Limb Effect. We found that participants generally attributed more responsibility to human assistants than to AI assistants, with responsibility shifting onto the assistant especially when the assistant underperformed or received negative feedback. Under high competence, human assistants were held more responsible than AI assistants, whereas under low competence this human–AI difference disappeared. However, when considering whether client feedback was “also about me,” a paradoxical asymmetry emerged only for AI: participants felt negative feedback about an AI as more self-directed, while positive feedback transferred far less; no such valence effect appeared for human delegates.

This contrast can be explained by distinguishing responsibility from accountability. Responsibility concerns who caused an outcome, whereas accountability reflects whether one is perceived as answerable in the eyes of others. Participants recognized that the AI assistant was responsible for the task outcome, but because AI is not human and cannot be fully held accountable in a social sense, they simultaneously felt that negative feedback toward the AI was also directed at them. In other words, accountability here emerges through perspective-taking [31]: individuals imagine how a client judges both themselves and their assistant, and in doing so, they absorb criticism that might otherwise fall solely on the AI. This finding aligns with [90, 98], who showed that when a moral violation occurs, humans are generally attributed greater moral fault than AI agents.

The AI phantom limb metaphor illustrates this asymmetry. A physical phantom limb often registers pain but rarely comfort. Similarly, participants “felt the pain” of criticism when their AI assistant was judged poorly, but they did not experience the “pleasure” of praise to the same degree. We therefore describe AI as a kind of

cognitive phantom limb—an extension of human intellectual capacity. Just as a phantom limb extends one’s body schema beyond its physical limits, AI extends cognitive agency. When that extension fails or is criticized, people feel the sting as if it were their own. When it succeeds, however, the compliment is deflected back to the AI rather than fully shared.

One theoretically plausible account for the observed effect can be derived from mind perception theory. Mind is typically conceptualized as comprising two dimensions—agency and experience [37]. While advancements in AI have substantially increased perceived agency [50], AI systems continue to be attributed low levels of experiential capacity or to be considered cryptomind entities [90]. When individuals are wronged or victimized, they tend to search for a responsible agent or scapegoat to blame [101]. This framework readily explains why participants generally attributed more responsibility to mindful human assistants than to AI assistants, with responsibility shifting toward the assistant particularly when the assistant underperformed or received negative feedback. However, when examining the direction of feedback, the dynamics become more intricate. When the assistant is an AI, individuals not only find it more difficult to attribute blame to an entity perceived as mindless; they also anticipate that a mindful evaluator (e.g., the client) holds the same belief—that AI lacks mental ability. This dual constraint reduces the plausibility of attributing blame to the AI and instead leads individuals to interpret negative feedback as personally targeted.

A further layer of complexity emerges when asking why are individuals more inclined to internalize negative feedback rather than positive feedback? Although the present study does not investigate the underlying psychological mechanisms, recent HCI research offers potential explanations for this phantom limb effect. In everyday contexts, AI use is often surrounded by stigmatizing slurs—for instance, using “AI could have written this” to undermine someone’s writing ability, or assuming that essays by non-native English speakers must be AI-generated [86]. Such practices help establish prevailing social norms that delegitimize or disparage AI-mediated work [18]. These external negative judgments can lead individuals to fear that their own capabilities will be questioned, prompting them to hide their use of AI in professional or academic settings [105]. In addition, AI use is often accompanied by internal negative judgments, such as questioning one’s own competence [105]—akin to imposter syndrome [25, 86]. Taken together, individuals may not only infer the client’s perspective within a single interaction, but also absorb broader societal slurs surrounding AI use and their own anxieties about relying on AI—both of which might further shape the internalization of negative feedback.

To sum up, these findings suggest that human–AI delegation reshapes the self–other boundary in ways distinct from human–human interaction. Importantly, this should not be conflated with the broader idea of the extension of self. Prior work shows that some users—especially in collectivist cultures—perceive AI as part of the self, treating its features as tools for conformity, social adaptation, or privacy protection [10]. If AI were truly functioning as such an extension, one would expect both praise and blame to transfer symmetrically. Instead, our results reveal a sharper asymmetry: participants deflected credit to the AI but internalized criticism

themselves. This is why we call it a phantom limb effect—AI operates as a cognitive extension that can transmit pain but not pleasure.

6.2.2 *The Distinction Between Responsibility and Accountability.*

The distinction between responsibility and accountability in our study clarifies the present findings and helps reconcile competing interpretations of the “moral crumple zone.” Elish [30], who first introduced the concept, described it as the misattribution of responsibility to a human actor with limited control over an automated system. By contrast, Hohenstein’s [43] empirical research suggested that AI itself can function as a moral crumple zone by absorbing responsibility that would otherwise have been assigned to the human. Our framework, which separates responsibility from accountability, offers a way to integrate these perspectives. Elish [30] illustrated her argument through technological disasters such as the Three Mile Island nuclear accident and the crash of Air France Flight 447, emphasizing that at the societal level, humans are typically regarded as the primary agents to be held accountable. Similarly, when adopting a third-person perspective in which they are not directly involved, observers tend to attribute greater moral blame to humans than to AI systems [90]. This perspective aligns with the asymmetries we observed in participants’ self-directed interpretations of feedback. Hohenstein [43], in turn, examined responsibility in AI-mediated communication, showing that failed interactions often shifted responsibility onto AI, reducing the burden placed on human partners. This dynamic resonates with our findings: participants frequently displaced responsibility onto assistants, especially when performance was poor or feedback was negative. Our experimental design therefore bridges these two viewpoints by assessing participants’ subjective responsibility attributions while they are directly engaged in the task, and by interpreting the client’s feedback as a reflection of accountability at the societal level.

Interestingly, our study is not the first to differentiate individuals’ subjective judgments from cognition shaped by internalized social expectations. Draxler et al.’s work [26] on the “AI ghostwriter” phenomenon demonstrates a similar divide: although participants often did not feel psychological ownership of AI-generated text, they nevertheless publicly declared themselves as its authors. This distinction between perceived ownership and declared authorship parallels our analytical focus. The former reflects the internal, subjective sense of responsibility—analogue to our construct of internal responsibility attribution—whereas the latter concerns outward-facing behavior that is embedded in social norms and the imagined presence of evaluative others. Human–AI collaboration thus requires navigating both internal perceptions and external social expectations. Recognizing this dual perspective provides a more comprehensive account of how individuals interpret feedback, make interpersonal judgments, and internalize broader normative frameworks surrounding AI use.

Taken together, these insights highlight the value of carefully distinguishing responsibility from accountability in future studies of moral crumple zones. Attribution in human–AI collaboration is complex and multifaceted, and conflating these concepts risks obscuring the subtle ways in which humans and AI are positioned within social and moral frameworks. By separating the two, researchers can more precisely capture how accountability

is negotiated and how responsibility is shared, ultimately advancing a clearer and more constructive understanding of human–AI collaboration.

6.3 Social and Design Implications

Our findings suggest that responsibility and accountability are created in distinctive and sometimes asymmetric ways when delegation involves AI. Socially, the phantom limb effect points to a troubling imbalance: positive outcomes are often credited to AI, leaving users feeling their own competence diminished, while negative outcomes are perceived as directed at the human, reinforcing a sense of blame. Such dynamics might risk creating unhealthy work environments where employees' contributions are undervalued and their failures over-exposed. Similar concerns have already been observed in education, where younger students report that their achievements are discounted as products of AI rather than their own effort [25]. Meanwhile, external critics warn that students' foundational knowledge and core academic competencies are weakening, leading employers to doubt the reliability of academic credentials and adopt "anti-AI" screening measures [20]. Traditionally, academic credentials have functioned as signals of graduates' underlying ability and productivity [28]. However, when these accomplishments are instead attributed to AI, the technology becomes a phantom limb that cannot share success or pride—credit flows to the tool rather than the individual. Such dynamics may intensify doubt toward one's own competence, reinforce impostor syndrome [25], and potentially contribute to broader post-graduation disillusionment with career prospect [41, 78]. In organizational contexts, this asymmetry could distort workplace hierarchies: clients or managers may implicitly trust AI for successes while holding employees accountable for failures, undermining both morale and professional recognition. This further explains not only why individuals hide AI use [45, 105], but also why cultural norms and even regulatory efforts emerge to stigmatize AI reliance [86]. Rather than tighter restrictions, a more constructive path lies in shifting norms around attribution and recognition. Organizations could explicitly reward effective human–AI collaboration as a form of competency—not a dilution of skill—while ensuring that responsibility and credit are shared transparently. Such cultural realignment may reduce the need for concealment, alleviate stigma, and allow the psychological benefits of collaboration to be fully realized.

Beyond cultural and organizational interventions, our findings also carry direct implications for the design of AI delegation systems. Design implications follow directly from these social risks. For delegation with AI is particularly sensitive to issues of responsibility and accountability, systems must be designed to clarify—not obscure—the boundaries of attribution. One potential design consideration is explainability: delegation interfaces could make explicit how outcomes are generated, when errors stem from contextual factors, and where responsibility resides. This means not only minimizing cognitive load but also providing users with transparent cues that prevent over-attribution of blame to themselves and over-crediting of success to AI. Concretely, delegation systems should (1) signal the provenance of outcomes and errors, (2) frame results in ways that recognize the user's role, and (3) ensure that accountability is distributed fairly across human and AI partners.

Taken together, these social and design implications emphasize the need to account for the asymmetrical ways in which people process responsibility and accountability in Human–AI collaboration. Without careful design, AI may function as both scapegoat and competitor—absorbing blame selectively while monopolizing credit—creating environments that are both psychologically unhealthy and socially inequitable. With thoughtful attention to cultural realignment, explainability and attribution, however, AI delegation systems can instead support trust, preserve recognition for human agency, and foster more sustainable forms of collaboration.

6.4 Limitation and Future Work

This study has several limitations that should be considered when interpreting its findings. First, the experiment was conducted entirely online in a highly simplified delegation scenario involving only a participant, an assistant, and a client. The interaction itself was pre-scripted and minimal: participants observed animated figures without real faces, histories, or contextual cues that would normally shape perceptions in everyday delegation. This abstraction limited ecological validity. In particular, the competence manipulation was designed such that high-competence assistants always secured mutually ideal time slots for both the client and the participant. In real-world collaboration, however, competence often involves trade-offs—an effective assistant might satisfy one party while imposing costs on another. Such complexities, which are central to workplace delegation, were not captured here. Future work should therefore develop richer, context-aware paradigms that more closely simulate the nuanced and dynamic nature of human–AI delegation.

Second, our sample was restricted to English-speaking participants, largely from individualist cultures. This limits the generalizability of our findings, as cultural differences in self–other perception, responsibility attribution, and expectations of technology may produce different dynamics in collectivist contexts. Longitudinal studies, in which participants interact with AI assistants repeatedly over time, would also offer deeper insights into how trust, accountability, and responsibility evolve as relationships with AI systems mature.

For future work, besides addressing the methodological considerations mentioned above, our results raise theoretical challenges that warrants future investigation. In particular, psychological distance—a construct often used to explain willingness to delegate—appeared highly aligned with trust in our data but failed to capture the more subtle blurring of self–other boundaries observed here. For example, when participants felt that negative feedback toward the AI was also directed at themselves, this could be interpreted as reduced psychological distance, yet such nuance is not reflected in existing measures that emphasize similarity or familiarity. Future research should therefore revisit and refine the concept of psychological distance in delegation contexts, integrating it more explicitly with accountability and responsibility attribution. In addition, future work may further investigate whether the internalization of negative feedback is driven by mechanisms related to mind perception as well as perceived negative external norms or judgments surrounding AI use.

From a design and societal perspective, our findings also highlight that delegation with AI is not only a technical challenge but

also a social one. Individuals may misperceive their relationship with AI delegates, internalizing negative feedback as self-directed while deflecting positive feedback to the system. These dynamic risks eroding human recognition, exacerbating blame, and creating unhealthy dynamics. To address these risks, future systems must move beyond efficiency to actively clarify accountability boundaries. Transparent, explainable interfaces that fairly recognize and distinguish both human's and AI's responsibilities and contributions are essential.

At the same time, these challenges open a pivotal opportunity for innovation. By anticipating how responsibility and accountability are shaped in Human–AI collaboration, researchers and designers can move beyond merely correcting misattribution to intentionally building systems that foster trust, strengthen human agency, and establish healthier collaboration dynamics. With thoughtful design, AI delegates can empower rather than be scapegoats or competitors. We are optimistic that by integrating these insights, the HCI and HAI communities can develop more human-centered delegation interactions, which shapes a future in which Human–AI collaboration genuinely empowers people.

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A Appendices

A.1 MANOVA with four subscales of trust scale

Table 5: MANOVA table regarding trust subscales and main independent variables

Main Effects		Value	Num DF	Den DF	F Value	Pr > F
Intercept	Wilks' lambda	0.58	4	344	62.265	<.001***
	Pillai's trace	0.42	4	344	62.265	<.001***
	Hotelling-Lawley trace	0.724	4	344	62.265	<.001***
	Roy's greatest root	0.724	4	344	62.265	<.001***
Feedback valence	Wilks' lambda	0.911	4	344	8.406	<.001***
	Pillai's trace	0.089	4	344	8.406	<.001***
	Hotelling-Lawley trace	0.097	4	344	8.406	<.001***
	Roy's greatest root	0.097	4	344	8.406	<.001***
Competence	Wilks' lambda	0.788	4	344	23.049	<.001***
	Pillai's trace	0.211	4	344	23.049	<.001***
	Hotelling-Lawley trace	0.268	4	344	23.049	<.001***
	Roy's greatest root	0.268	4	344	23.049	<.001***
Assistant type	Wilks' lambda	0.988	4	344	1.039	0.386
	Pillai's trace	0.011	4	344	1.039	0.386
	Hotelling-Lawley trace	0.012	4	344	1.039	0.386
	Roy's greatest root	0.012	4	344	1.039	0.386

Table 6: MANOVA of Perceived Risk subscale

Interaction Effects		Value	Num DF	Den DF	F Value	Pr > F
Feedback X Competence	Wilks' lambda	0.983	4	344	1.448	0.217
	Pillai's trace	0.016	4	344	1.448	0.217
	Hotelling-Lawley trace	0.016	4	344	1.448	0.217
	Roy's greatest root	0.016	4	344	1.448	0.217
Feedback X Assistant type	Wilks' lambda	0.980	4	344	1.679	0.154
	Pillai's trace	0.019	4	344	1.679	0.154
	Hotelling-Lawley trace	0.019	4	344	1.679	0.154
	Roy's greatest root	0.019	4	344	1.679	0.154
Competence X Assistant type	Wilks' lambda	0.964	4	344	3.213	0.013*
	Pillai's trace	0.036	4	344	3.213	0.013*
	Hotelling-Lawley trace	0.037	4	344	3.213	0.013*
	Roy's greatest root	0.037	4	344	3.213	0.013*
Feedback Valence X Competence X Assistant type	Wilks' lambda	0.983	4	344	1.465	0.212
	Pillai's trace	0.016	4	344	1.465	0.212
	Hotelling-Lawley trace	0.017	4	344	1.465	0.212
	Roy's greatest root	0.017	4	344	1.465	0.212

Table 7: MANOVA of Benevolence subscale

	SS	DF	MS	F	p-value	η_p^2
Assistant Type	18.420	1.0	18.420	21.397	<.001***	0.180
Competence level	59.949	1.0	59.979	69.998	<.001***	0.167
Feedback valence	65.707	1.0	65.707	76.683	<.001***	0.058
Assistant type X Competence	8.851	1.0	8.851	10.330	0.001**	0.002
Assistant type X Feedback valence	0.454	1.0	0.454	0.530	0.466	0.001
Competence X Feedback valence	0.813	1.0	0.813	0.949	0.330	0.028
Assistant Type X Competence X Feedback valence	0.000	1.0	0.000	0.000	0.990	0.000

Table 8: MANOVA of Competence subscale

	SS	DF	MS	F	p-value	η_p^2
Assistant Type	18.420	1.0	18.420	21.397	<.001***	0.180
Competence level	59.949	1.0	59.979	69.998	<.001***	0.167
Feedback valence	65.707	1.0	65.707	76.683	<.001***	0.058
Assistant type X Competence	8.851	1.0	8.851	10.330	0.001**	0.002
Assistant type X Feedback valence	0.454	1.0	0.454	0.530	0.466	0.001
Competence X Feedback valence	0.813	1.0	0.813	0.949	0.330	0.028
Assistant Type X Competence X Feedback valence	0.000	1.0	0.000	0.000	0.990	0.000

Table 9: MANOVA of Reciprocity subscale

	SS	DF	MS	F	p-value	η_p^2
Assistant Type	18.420	1.0	18.420	21.397	<.001***	0.180
Competence level	59.949	1.0	59.979	69.998	<.001***	0.167
Feedback valence	65.707	1.0	65.707	76.683	<.001***	0.058
Assistant type X Competence	8.851	1.0	8.851	10.330	0.001***	0.002
Assistant type X Feedback valence	0.454	1.0	0.454	0.530	0.466	0.001
Competence X Feedback valence	0.813	1.0	0.813	0.949	0.330	0.028
Assistant Type X Competence X Feedback valence	0.000	1.0	0.000	0.000	0.990	0.000

A.2 ANOVA on four subscales of trust scale

Table 10: Three-way ANOVA of Perceived Risk (Left) and Benevolence (Right)

	Perceived Risk				Benevolence			
	DF	F	p-value	η_p^2	DF	F	p-value	η_p^2
Assistant Type	1	21.497	<.001***	0.040	1	22.957	<.001***	0.038
Competence level	1	69.998	<.001***	0.1328	1	175.736	<.001***	0.295
Feedback valence	1	76.683	<.001***	0.1455	1	44.961	<.001***	0.075
Assistant type X Competence	1	10.330	0.001**	0.019	1	0.289	0.591	0.000
Assistant type X Feedback valence	1	0.530	0.466	0.001	1	2.828	0.093	0.004
Competence X Feedback valence	1	0.949	0.330	0.001	1	1.895	0.169	0.003
Assistant Type X Competence X Feedback valence	1	0.000	0.990	0.000	1	0.0128	0.910	0.000

Table 11: Means and standard deviations of four trust subscales by feedback valence, competence level, and assistant type

Factor	Level	Perceived Risk Mean (SD)	Benevolence Mean (SD)	Competence Mean (SD)	Reciprocity Mean (SD)
Feedback Valence	Positive	3.13 (1.06)	3.48(1.15)	3.43 (1.31)	3.18 (1.21)
	Negative	2.27 (1.03)	2.82 (1.18)	2.42 (1.32)	2.38 (1.15)
Competence level	High	3.11 (1.10)	3.81 (0.92)	3.84 (1.00)	3.52 (1.00)
	Low	2.29 (1.00)	2.50 (1.11)	2.04 (1.15)	2.06 (1.02)
Assistant Type	AI	2.47 (1.02)	2.92 (1.13)	2.78 (1.33)	2.57 (1.13)
	Human	2.92 (1.19)	3.38 (1.25)	3.07 (1.47)	2.99(1.32)

Table 12: Three-way ANOVA of Perceived Risk (Left) and Benevolence (Right)

	SS	DF	MS	F	p-value	η_p^2
Assistant Type	18.420	1.0	18.420	21.397	<.001***	0.180
Competence level	59.949	1.0	59.979	69.998	<.001***	0.167
Feedback valence	65.707	1.0	65.707	76.683	<.001***	0.058
Assistant type X Competence	8.851	1.0	8.851	10.330	0.001**	0.002
Assistant type X Feedback valence	0.454	1.0	0.454	0.530	0.466	0.001
Competence X Feedback valence	0.813	1.0	0.813	0.949	0.330	0.028
Assistant Type X Competence X Feedback valence	0.000	1.0	0.000	0.000	0.990	0.000

Table 13: Three-way ANOVA of Perceived Risk (Left) and Benevolence (Right)

	SS	DF	MS	F	p-value	η_p^2
Assistant Type	18.420	1.0	18.420	21.397	<.001***	0.180
Competence level	59.949	1.0	59.979	69.998	<.001***	0.167
Feedback valence	65.707	1.0	65.707	76.683	<.001***	0.058
Assistant type X Competence	8.851	1.0	8.851	10.330	0.001**	0.002
Assistant type X Feedback valence	0.454	1.0	0.454	0.530	0.466	0.001
Competence X Feedback valence	0.813	1.0	0.813	0.949	0.330	0.028
Assistant Type X Competence X Feedback valence	0.000	1.0	0.000	0.000	0.990	0.000

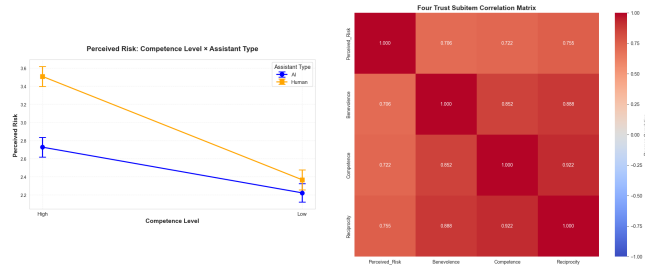


Figure 5: Interaction plot illustrating the two-way interaction between High vs. Low condition and AI vs. Human assistant on Perceived Risk (± 1 SE). The Human condition shows a markedly steeper decline from High to Low, decreasing from $M = 3.51$ ($SE = 0.11$) to $M = 2.37$ ($SE = 0.11$), whereas the AI condition exhibits a comparatively flatter slope, shifting from $M = 2.73$ ($SE = 0.11$) to $M = 2.22$ ($SE = 0.10$). (Right) Correlation matrix of the four trust sub-scales—Perceived Risk, Benevolence, Competence, and Reciprocity—demonstrating consistently strong positive associations across them.

A.3 Responsibility Attribution to the Client

Table 14: Three-way ANOVA of Responsibility Attribution to the Client

	Attribution to Client		
	F	p-value	η_p^2
Assistant Type	1.274	0.259	0.003
Competence level	17.137	<.001***	0.047
Feedback Valence	5.531	<0.05**	0.015
Assistant type X Competence	11.095	<.001***	0.030
Assistant type X Feedback valence	0.649	0.420	0.001
Competence X Feedback valence	16.761	0.000	0.004
Assistant Type X Competence X Feedback Valence	0.445	0.504	0.001